

Magneto-Optical Kerr Effect and Magnetic Anisotropies(MOKE)

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1 Introduction

This experiment focuses on the magneto-optical Kerr-effect (MOKE) which is used to determine the properties of a ferromagnet by measuring of it's optical polarization. We will be measuring on a longitudinal configuration, one of the three possible geometrical configurations that depend on the magnets orientation (along with polar and transversal configurations; see fig.1a).MOKE signals are interesting because of their proportionality to the sample magnetization which allow us to determine the MOKE hysteresis loops and physical quantities such as, coercitivity, remanence and magnetic anisotropy.For this experiment we will look into a thin BCC-Fe (001) single crystal and predict its shape and crystal anisotropy by measuring the hard and easy magnetization axis and we will use a polar Kerr microscope to visualize the domain pattern of a rare-earth iron garnet film and to examine the behaviour of the domains in external magnetic fields

2 Theoretical Background

2.1 Ferromagnetism and Domains

A ferromagnetic sample is remanence builds up a magnetic pattern and it's determined by a combination of exchange interaction, stray field energy,crystal anisotropy and zeemann energy. Materials which has net magnetic moments is called ferromagnetic material. It consists of smaller areas is named domain and each domain magnetic moments are aligned in certain direction which is shown in figure 1.

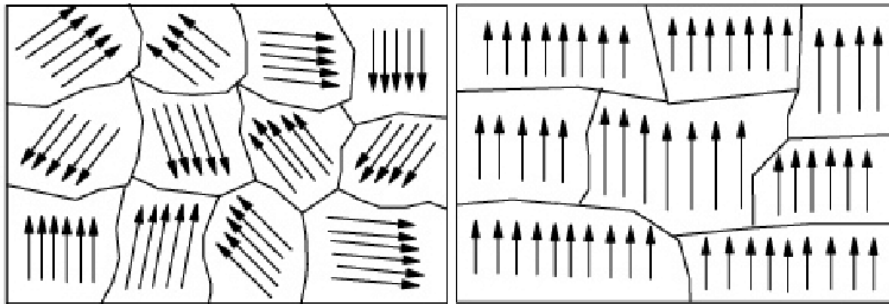


Figure 1: Graphical representation of magnetic moments and domain walls[4]

This lined up is due to the exchange interaction (spin inter-action)depend on atomic

distance. Every domain got some net magnetization in certain direction and it varies domain to domain and net magnetization posses zero. When applying external magnetic field, all the domain moments point towards a certain direction giving rise to spontaneous magnetization.

2.2 MOKE Geometry

There are there different types of geometries depending on magnetization and light incidence is used to investigate the magnetic properties of thin film. These are polar , longitudinal and transverse . In case of polar, magnetization is perpendicular to the sample surface and for longitudinal, Magnetization is parallel to sample surface as well as for transverse term, magnetization is perpendicular to the plane of incidence and parallel to surface. The following figure(2) depicts the there experimental setup.

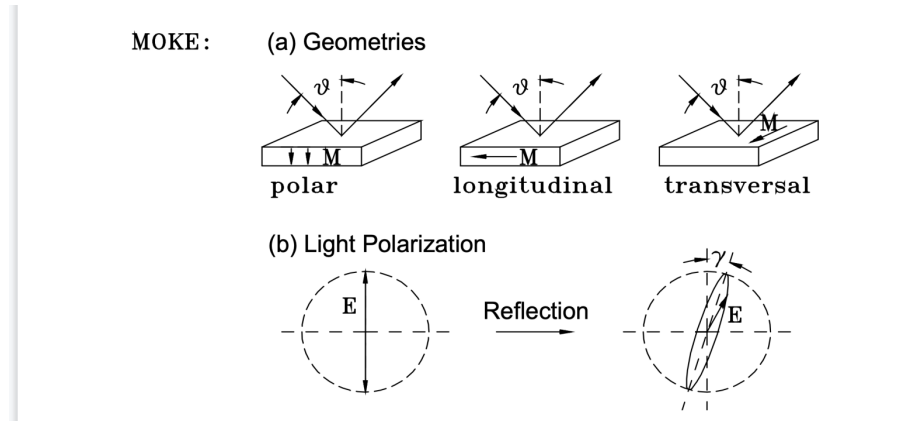


Figure 2: Schematic representation of MOKE geometry and light polarization[4]

2.3 Magneto-optical Kerr effect

As it name implies, a magneto-optical effect occurs when light is transmitted through magnetized matter and then subsequently reflected from a magnetized surface. Depending on the polarization of the incoming light and the magnetization of the sample, a change of the direction of polarization and an ellipticity of the transmitted or reflected light is observed (see Fig. 2b). If a linearly polarized light bounces off from a metallic surface, it retrieve its polarization again. On the other hand, this mechanism doesn't hold

for magnetized surface and the state of polarization changes. The changes of polarization state can be describe by using dielectric tensor as follow $\mathcal{E}_l = \mathcal{E}_0 Q m_i$ where Q is the voigt magneto optical constant and is the ratio of magnetization in a given direction and the magnitude of saturation magnetization of the sample $m_i = \frac{M_i}{M_{Satu}}$. The off diagonal term of the dielectric tensor are proportional to the magnetization, changes the polarization state of light relative to magnetization of the sample. This magneto optical response which is measured in this experiment since the rotation of the polarization with relative to incident light.

The complex Kerr rotation for a s-polarized light can be obtained by:

$$\Theta_k^s = \frac{-r_{ps}}{r_{ss}} \quad (1)$$

And for p-polarized light

$$\Theta_k^p = \frac{-r_{sp}}{r_{pp}} \quad (2)$$

2.4 Magnetic Hysteresis

Depending on the shape induced demagnetization field and easy and hard axes a thin sample is usually not magnetized homogeneously. Describing the hysteresis loops quantitatively is difficult, but it is possible to make a rough estimate for the anisotropy constant K_1 from the difference between hard and easy axis hysteresis loops. When all the magnetic

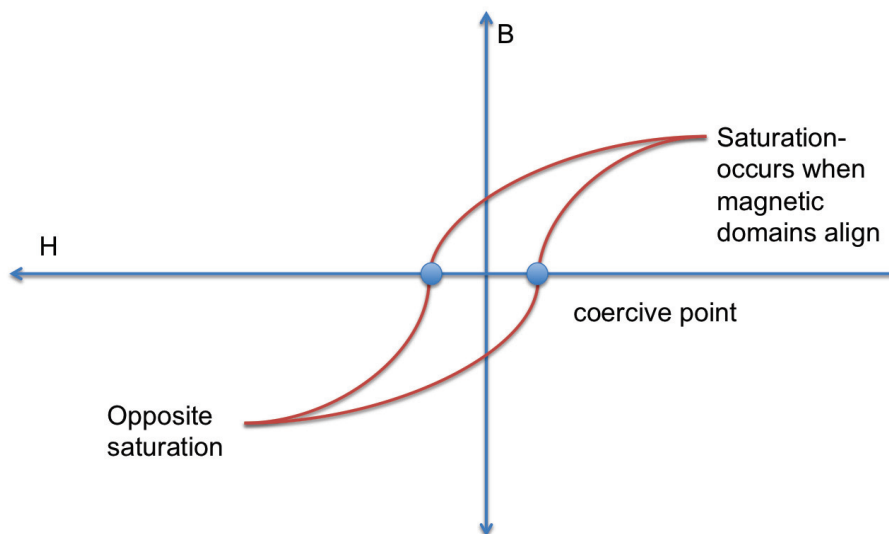


Figure 3: Schematic representation of MOKE geometry and light polarization[4]

moments aligned within a certain material in same direction, the magnetic permeability takes a constant value. At this point, the magnetization will reach a saturation point where it show maximum or minimum value. The point where magnetization get zero value is called coercive point that depicts how a magnetic material is demagnetization when expose to external magnetic field. By measuring the changes of coercivity relative to incidence angle, it's easy to describes the amount of energy needed to change the magnetization in a particular direction for a specific sample in external magnetic field. When the saturation point takes negative values, it's called negative saturation.

2.5 Magnetic Anisotropy

Because of the nature of dipole-dipole interaction and of spin-orbital coupling there are preferred directions for the spontaneous magnetization inside a crystal. The direction with lowest energy necessary in bcc Fe is the [100] direction, this is called easy axes. The [110] and the [111] directions on the other hand are called hard axes. The energies needed per volume for the spontaneous magnetization along different crystal axes are described by the crystal anisotropy $E - cryst$. The shape anisotropy E_{shape} , on the other hand describes the magnetic stray field energy per volume as a function of the magnetization direction, it depends therefore on the shape of the magnetic sample.

2.6 Anisotropy of Energy

As for the shape of sample is concerned it looks like an ellipsoid which leads to anisotropy in energy and can be written as:

$$E_{shape} = \frac{\mu_0}{2} H_D M_S = \frac{\mu_0}{2} N M_s^2 \quad (3)$$

Also it can be calculated using the following equation:

$$\frac{E}{V} = K_0 + K_1(\alpha^2\beta^2 + \beta^2\gamma^2 + \alpha^2\gamma^2) + K_2\alpha^2\beta^2\gamma^2 \quad (4)$$

where K_i are the anisotropy constants and α, β, γ are the direction cosines.

2.7 Magneto optics

The possible effects of the magnetization acting on electromagnetic waves like light are concisely describe by Magneto Optical Kerr effect. Observed effects are Faraday effect

and Kerr effect. The both effect describe the same phenomena based on same theory. The polarization state of light changes when passing through a electromagnetic field. Faraday effect depicts the changes of polarization while transmission through a material, on the other hand Kerr effect explain

3 Experimental Setup

The MOKE experiment is done using He-Ne laser as a light source. The schematic diagram of the set-up is given in figure 1.

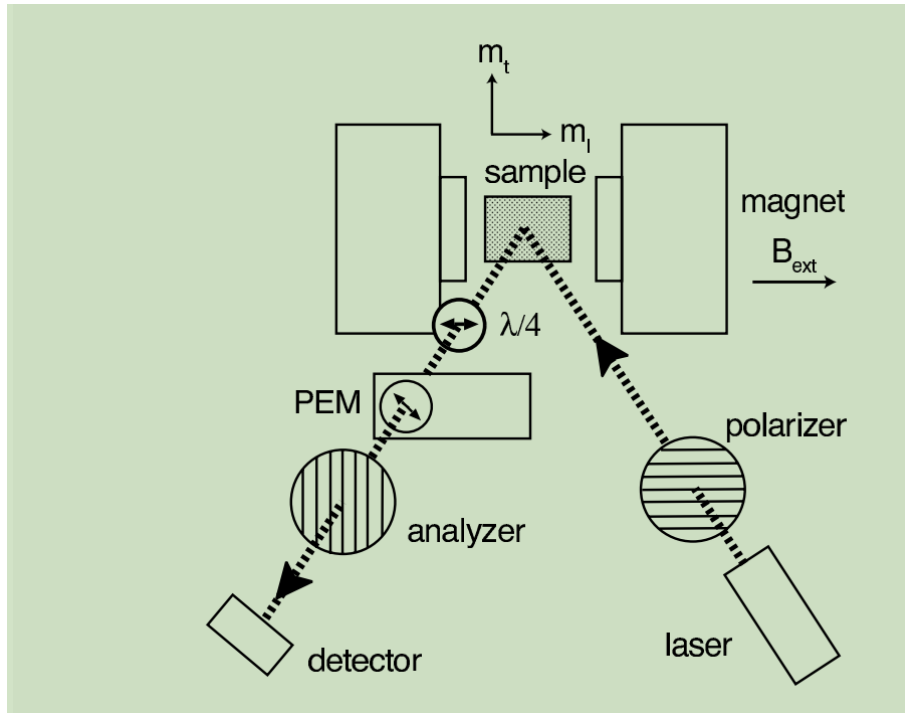


Figure 4: Schematic representation of MOKE geometry and light polarization[4]

A He-Ne laser of wavelength 670nm and FWHM is less than 2nm is used in MOKE. The laser beam is then passed through a polarizer to get the expected intensity as the laser is not a source of polarized light and projected on the sample which is placed in an in-plane magnetic field (Ring shaped iron core which is wrapped by a coil) produced by a broke electromagnet (Power- 0.7T). A rotatable holder with an angle measurement scale is used on which the sample is kept and it determines the angle with respect to the plane around $\pm 2^\circ$. The projected laser beam get reflected from the Fe thin film which then passes through a polarizer again and the intensity is then transformed into a electric

voltage. A diaphragm (Beam limiting apparatus) is used between the sample and the optical instrument to avoid multiple reflection and to have beam of narrow wide. Direct measurement of Kerr rotation or changes of refractive index is quite impossible because of having small changes of the direction($\pm 1^\circ$). So by rotating the analyser at regular interval, Kerr rotation has been recorded as a function of intensity versus rotation angle. This output signal is then amplified since it's a very weak signal and measured data are presented on a computer with digital interface.

4 Experimental Procedure

The experimental procedures are done as follows,

- The initialization of magneto optical Kerr effect is begin with measuring the strength of the electromagnet by using a hall sensor.
- Now mounting and adjusting the essential hardware components for the measurement of the MOKE (Adjustment of He-Ne laser, beam analyser, lock-in amplifier, Digital interface for data presentation and analysis)
- Observing the changes of polarization state of the reflected light from the sample by rotating the polarizer by 5 degree. Connection between observed reflected signals and rotation angle of the sample plane are done by modulator.
- The Kerr-ellipticity is determined by putting a plate ($\lambda/4$) in between the sample and PEM.
- The Kerr rotation and ellipticity is now measured for expected sample(Fe).

5 Kerr Microscope

The operation principle of Kerr microscope is same like Kerr magnetometer except the detector unit. A CCD camera is used instead of photo-diode and the modulator is omitted. In order to focus the sample surface on the CCD camera, it necessary to adjust the lenses of the the incident and reflected beam. Adjusting the analyzer for minimum intensity and rotating it until a double intensity for the minimum position is

obtained. The Kerr rotation is measured by increasing or decreasing the light intensity through the analyzer relative to rotation direction which also gives local magnetization direction of the sample. Different magnetization with dark or bright areas indicates magnetic domain. This can be seen easily by changing the power of external magnetic field.

6 Tasks

6.1 Calculate a theoretical approximation for the calibration factor in the center. For the theoretical derivation, use Ampere's law. Number of coil windings 300; slit width 12 mm).

$$\oint_c \mathbf{B} = \mu_0 N I \quad (5)$$

and we can write it as:

$$\oint_c \mathbf{B} = \frac{1}{\mu} (B_{\text{coil}}(2\pi r - s) + B_{\text{slits}}) = \mu_0 N I \quad (6)$$

so from here we get to the result

$$\mathbf{B} = \frac{\mu_0 N I}{s} \quad (7)$$

using values $N = 300$ and slit width $s = 12\text{mm}$, we can calculate $B = 31.4 \frac{\text{mT}}{\text{A}}$

6.2 Derive an expression for the crystalline magnetic anisotropy energy (per volume) as a function of angle between magnetization and crystal axes and anisotropy constants K_i ($i = 0, 1, 2$, etc.; to first order in K). Express the anisotropy constant K_1 in terms of the energy difference between the two in plane crystal directions.

Already done in section 2.6

6.3 Derive the appropriate formula before doing the experiment!

$$C = \frac{I_2 - I_1}{I_0} = 2\left(\frac{I_2 - I_1}{I_2 + I_1}\right) \quad (8)$$

Using Maulls law and small angle approximation we cane get to the result as:

$$2\frac{(\alpha + \phi)^2 - (\alpha - \phi)^2}{(\alpha + \phi)^2 + (\alpha - \phi)^2} = 4\frac{\phi\alpha}{\alpha^2 + \phi^2} \quad (9)$$

if we again consider α very much smaller than ϕ we get

$$C = 4\frac{\alpha}{\phi} \quad (10)$$

7 Result and Analysis

7.1 Calibration

In the experimental setup, the first step involved calibrating the magnetic field. This calibration was carried out by measuring the magnetic induction B at two locations: inside the center of the slit and at the position of the sample. To perform these measurements, a Hall probe was employed[1].

To obtain a theoretical approximation for the calibration factor in the center, we utilized Ampere's law. Ampere's law relates the magnetic field strength to the electric current flowing through a closed loop. By applying Ampere's law, we derived an equation to estimate the calibration factor for the magnetic field measurement at the center of the slit. This theoretical approximation allowed us to establish a relationship between the measured magnetic induction and the corresponding current values. This all calculations can be found in section 6.1.

$$B = 31.4 \frac{mTI}{A}$$

we will use this calibration value in determining the anisotropy constant K .

7.2 Magnetic Anisotropy

we investigate the magnetic anisotropy of a material by analyzing its hysteresis loops at various angles ranging from 45^0 to 180^0 degrees. The hysteresis loops were normalized for each angle using the Origin software, allowing for a comprehensive comparison of their behavior.

Through a meticulous process, we identified the easy and hard axes, crucial for understanding the material's magnetic properties. Notably, the saturation point at 120^0 degrees indicated a fast response, while the curvature towards saturation at 160^0 degrees suggested the presence of a hard axis. Using Python, we calculated the respective areas of the easy and hard axes:

enabling the determination of the magnetic anisotropy constant through a formula:

$$K = 4 \times (\text{Calibration factor})(A_h - A_e) \implies 4 \times 31.4(0.182 - 0.216) = -4.27 J/m^3 \quad (11)$$

The resulting magnetic anisotropy constant was found to be -4.27 constant signifies the

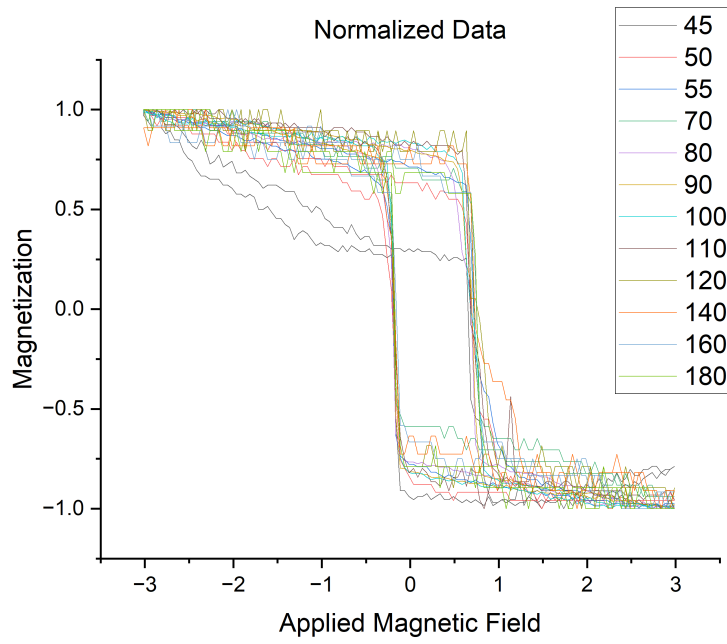


Figure 5: Normalized Data for different angles[4]

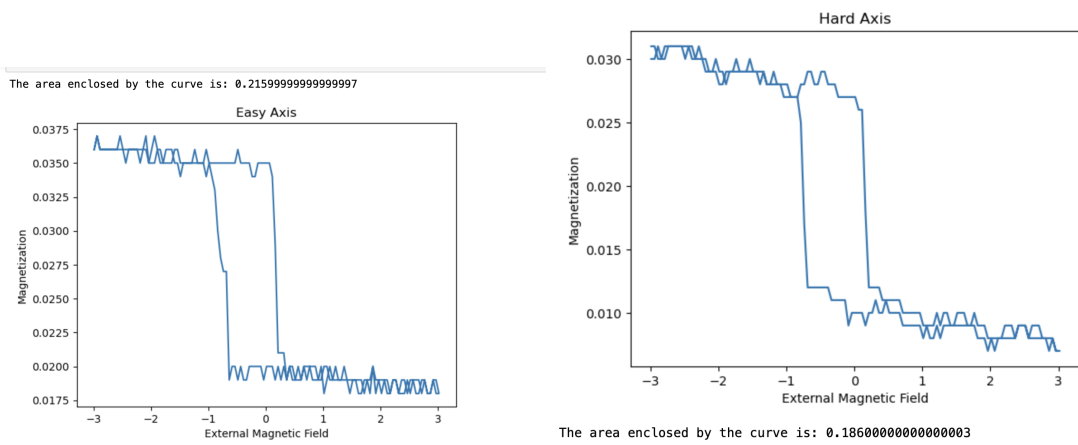


Figure 6: The Easy Axis (Left) and Hard Axis (Right) with their respective Areas

strength of magnetic alignment in a particular direction within the material. A higher value implies a strongly oriented alignment of magnetic dipoles, whereas a lower value suggests a less oriented and more random alignment, but here a negative value of the magnetic anisotropy constant means that the magnetic material prefers to have its magnetic moments oriented in directions different from those aligned with the crystallographic axes. This may result in more complex magnetic behavior and stability.

7.3 Kerr Angle

Direct measurement of the Kerr angle proved challenging due to its exceptionally small value. To overcome this limitation, we adopted an alternative approach to estimate the Kerr angle indirectly. Initially, the ferromagnetic sample's angle was fixed at 180° , and we systematically varied the incidence angle α for both s-polarized and p-polarized laser beams, with increments of $\pm 5^\circ$. Subsequently, we obtained the maximum and minimum values from the hysteresis graph at each angle setting.

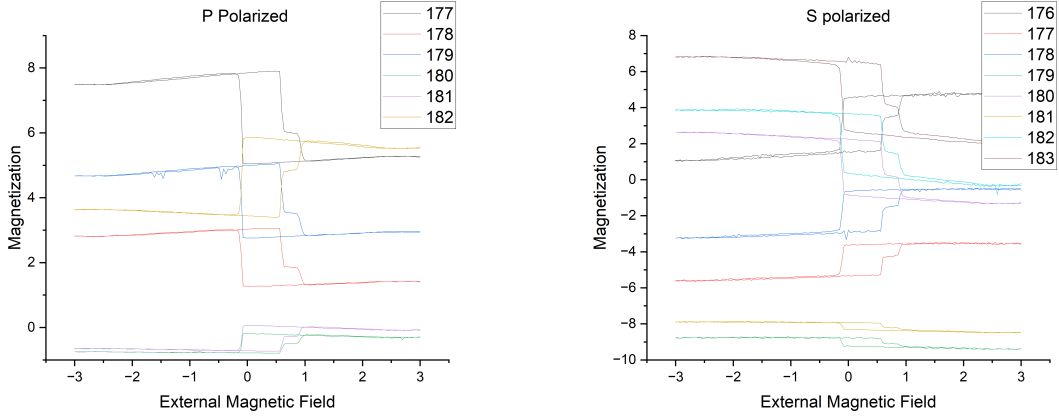


Figure 7: The P-Polarized (Left) and S-Polarized (Right) Hysteresis for $\alpha = \pm 5^\circ$

Leveraging these data, we computed the contrast using the formula derived from equation (8). This enabled us to determine the Kerr angle for both polarized beams as 0.071° for p-polarized laser and 0.045° for s-polarized laser. Our new approach successfully allowed us to derive meaningful values for the Kerr angle despite its minuscule magnitude. So, here is the contrast data plotted with angle of incidence from where we were able to calculate the Kerr angle:

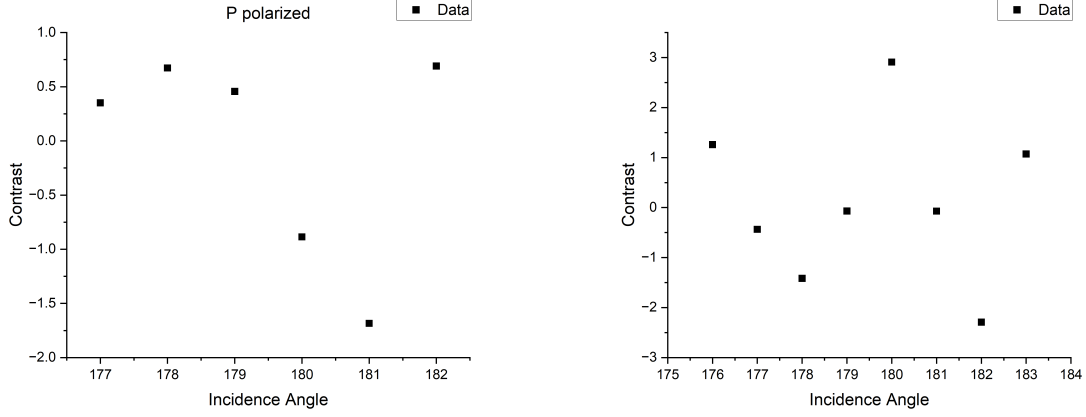


Figure 8: The P-Polarized (Left) and S-Polarized (Right) Contrast with respect to the angle of incidence and also $\alpha = \pm 5^\circ$

7.4 Kerr Microscopy

Kerr microscopy was employed to investigate the presence of distinct magnetic domains. This imaging technique operates based on the principle of illuminating a magnetized surface with polarized light and subsequently analyzing alterations in the polarization of the reflected light. By carefully examining and interpreting these polarization changes, the Kerr microscopy system generated visual representations of magnetic domains.

In accordance with theoretical expectations, the acquired images exhibited distinct characteristics. Brighter regions within the recorded images corresponded to domains characterized by spin up orientation, while darker regions corresponded to domains exhibiting spin down orientation. This contrast in brightness allowed for the identification and visualization of different magnetic domains within the sample under investigation.

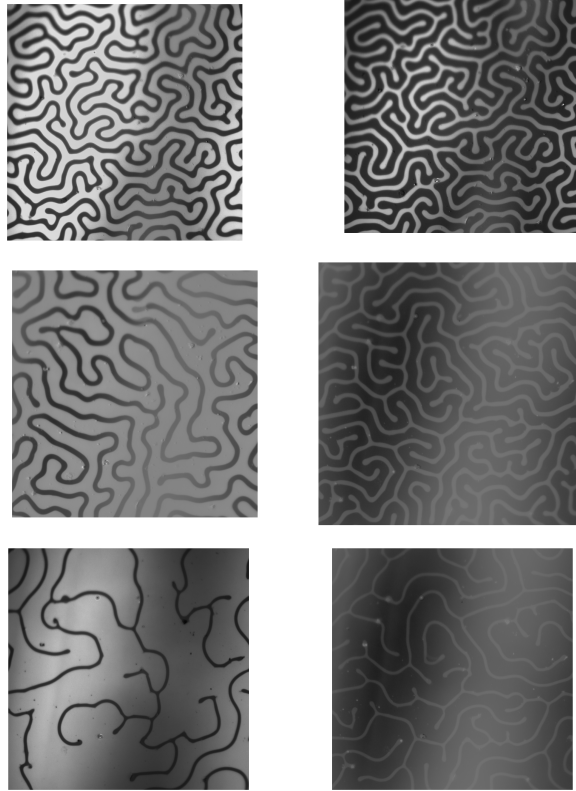


Figure 9: Different Kerr Microscopic images for Different Applied Magnetic Fields
 $-2021\mu T$ (Top Left) $2021\mu T$ (Top Right) $-3158\mu T$ (Middle Left) $3158\mu T$ (Middle Right)
 $-6538\mu T$ (Bottom Left) $5526\mu T$ (Bottom Right)

8 Conclusion

We conclude that the experiment focused on investigating the Magneto-optical Kerr effect, which demonstrates the influence of magnetization on the reflection of light from a magnetic material's thin film. Through the analysis of hysteresis loops, the easy and hard magnetization axes were determined, and the magnetic anisotropy constant was calculated. However, due to different units and the lack of quantitative comparisons with existing literature, the success of the obtained result remains uncertain. The examination of magnetic contrast and Kerr rotation revealed a small Kerr angle, which was obtained through the analysis of contrast using Malus's law. However, the absence of theoretical data for comparison leaves doubts about the accuracy of this result. The Kerr microscope images provided insights into the domain determination where bright colour defines the spin oriented up and the dark defines the spin oriented down. In summary, while the experiment yielded valuable observations regarding the Magneto-optical Kerr effect, uncertainties related to quantitative comparisons, theoretical data, and the accuracy of certain results highlight areas for further investigation and refinement in future studies.

References

- [1] [https://download.uni-mainz.de/physik-komet335/F-Praktikum/Script-V53-Moke Deutsch Version20090424.pdf](https://download.uni-mainz.de/physik-komet335/F-Praktikum/Script-V53-Moke_Deutsch_Version20090424.pdf)
- [2] <https://www.researchgate.net/figure/Experimental-MOKE-setup-The-quarter-wave-plate-is-introduced-to-measure-the-Kerr-fig1-1914703>
- [3] <https://www.researchgate.net/figure/Magnetic-domains-of-ferromagnetic-materials-in-the-absence-left-and-presence-right-fig2-290994683>.
- [4] [https://wiki.physik.fu-berlin.de/fp/private/ma12 magneto-optical kerr-effect and magnetic anisotropy](https://wiki.physik.fu-berlin.de/fp/private/ma12_magneto-optical_kerr-effect_and_magnetic_anisotropy).
- [5] [https://www.wm.edu/as/physics/documents/seniorstheses/class2016theses/heimburger matthew.pdf](https://www.wm.edu/as/physics/documents/seniorstheses/class2016theses/heimburger_matthew.pdf)
- [6] <https://www.researchgate.net/publication/51622689> Scanning magneto-optical Kerr microscope with auto-balanced detection scheme